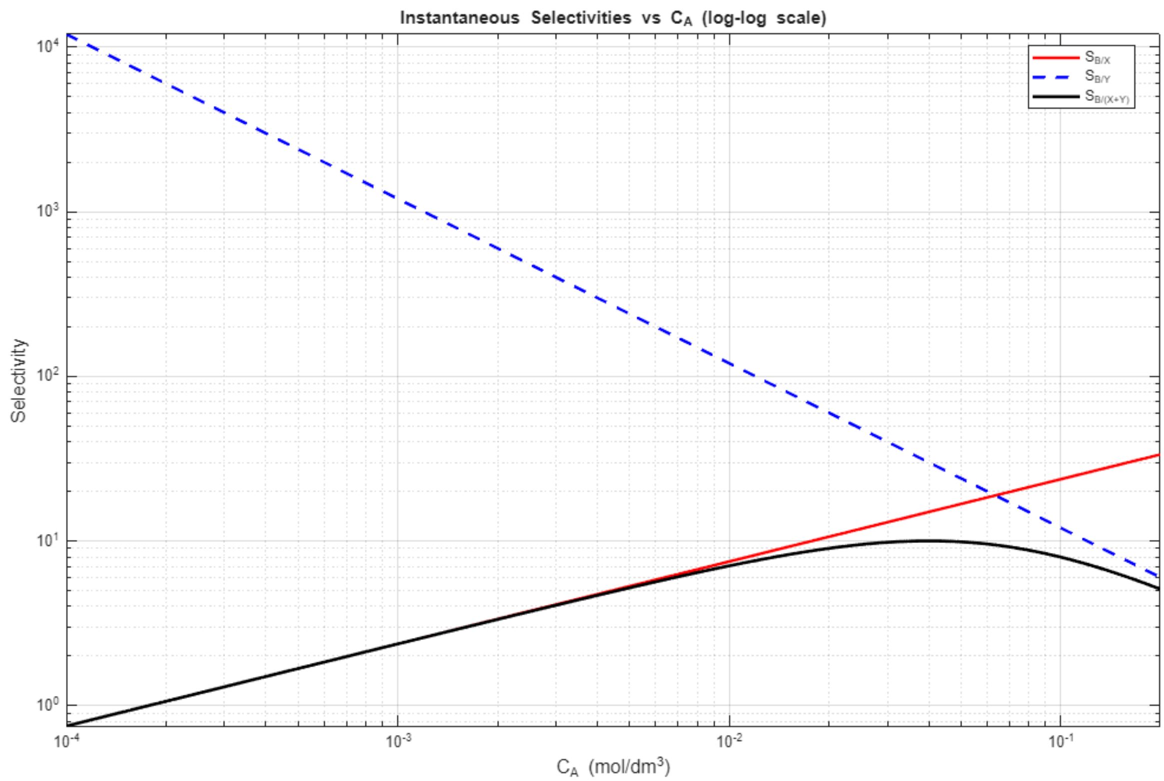
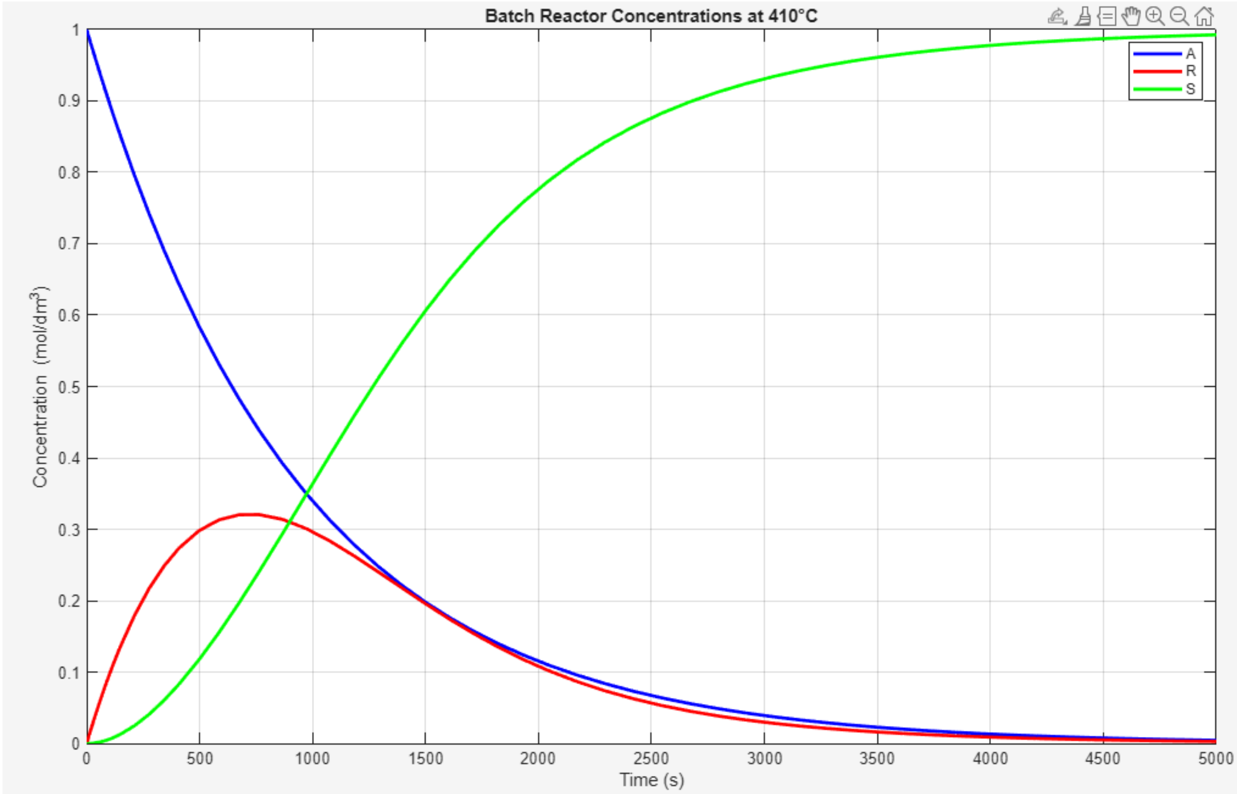
Problem 8-4



a)
b) 4.61 dm^3
c) Concentration of A: 0.0397, Concentration of B: 0.00548, Concentration of X: 0.000367, Concentration of Y: 0.000181
d) 60.28%
e) The reactor scheme you should use is a CSTR, with a volume of 4.61 dm^3, followed by a PFR, with a volume of 10.27 dm^3, in order to improve selectivity with 99% conversion of A.
f) 273 K
g) 1 atm

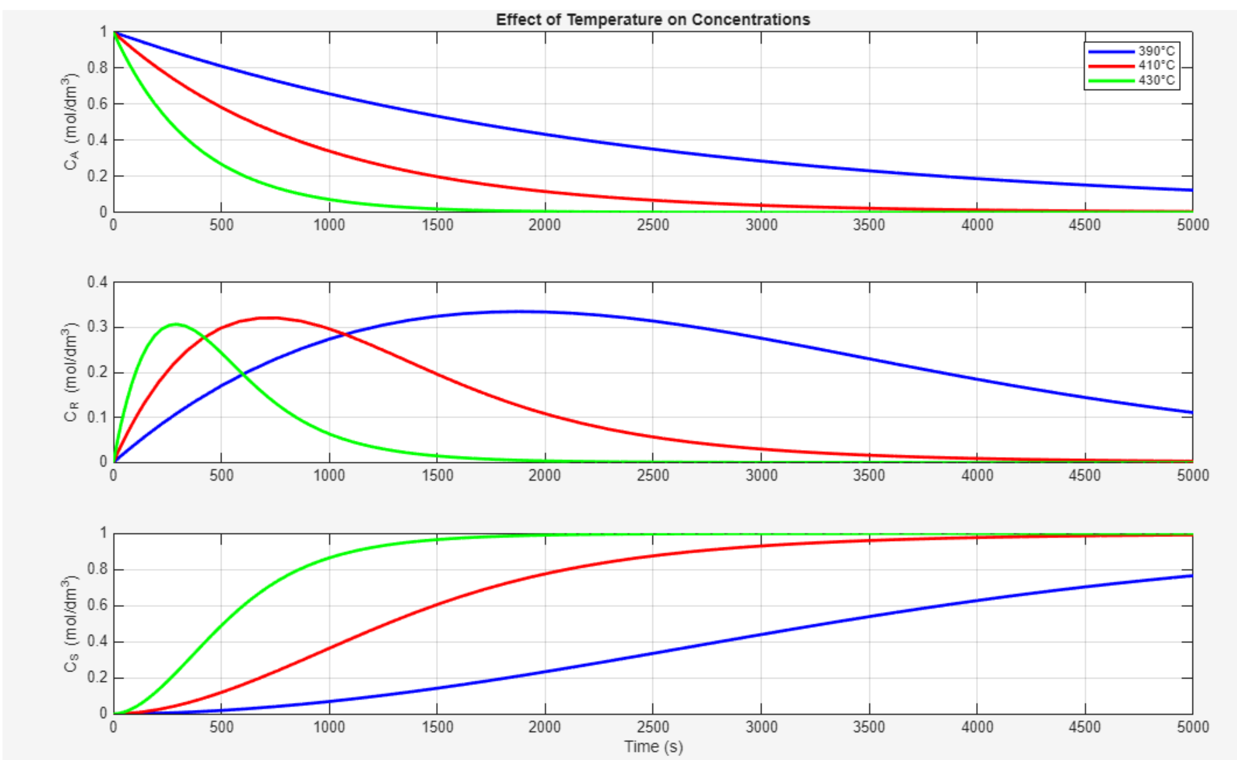
Problem 8-10

a)



At 390°C: Max R = 0.335 mol/dm^3 at t = 1874.4 s, At 410°C: Max R = 0.321 mol/dm^3 at t = 762.6 s, At 430°C: Max R = 0.307 mol/dm^3 at t = 292.6 s

b)



c) Ca = 0.4355 mol/dm^3, Cr = 0.1784 mol/dm^3, Cs = 0.3861 mol/dm^3